

SANDRA MÁRQUEZ DURÁN



MSc. in Statistics and Operations Research | Postgraduate in AI with Deep Learning | Licentiate in Mathematics.

Passionate about applying mathematics, statistics, and AI to solve real-world problems, with a focus on data science, machine learning, deep learning, and public health/clinical research.

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WORK EXPERIENCE

2018 Apr-Present	STATISTICAL TECHNICIAN at Barcelona Institute for Global Health (ISGlobal), Barcelona.
2017 Nov-2018 Mar	STATISTICAL TECHNICIAN at IDIBELL- Catalan Institute of Oncology L'Hospitalet de Llobregat, Barcelona
2017 Jun-Oct	STATISTICAL TECHNICIAN at Catalan Institute of Oncology L'Hospitalet de Llobregat, Barcelona.
2016 Oct-2017 Jun	INTERNSHIP AS STATISTICAL TECHNICIAN at Catalan Institute of Oncology L'Hospitalet de Llobregat, Barcelona.
2016-Mar-Ju	EDUCATIONAL CONTENT EDITOR at Epsilon Formación, Barcelona.
2015- Ap-Jul	DATA CONSULTANT at BaseTIS, Barcelona.

EDUCATION

Oct 2025 - March 2026 Postgraduate Course in Artificial Intelligence with Deep Learning UPC.

[Postgraduate course Artificial Intelligence with Deep Learning | UPC School - On Line](#)

Data Science training content: Deep Learning, Natural Language Processing, Computer Vision, Advanced Applications (Advanced Computer Vision 3D reconstruction, Reinforcement Learning), Final Project.

2016-Jun 2017 Support technician at Catalan Institute of Oncology. Collaboration with the department, Cancer Epidemiology Research Program (Cancer and Infections Unit) completing my master's thesis.

2015-2017 Master's degree in Statistics and Operations Research (MESIO), at Universitat Politècnica de Catalunya.

Pathway of interest: Biostatistics.

2015-2017	During these courses I acquired some general statistical knowledge and studied many subjects from my pathway of interest: • Epidemiology • Longitudinal Data Analysis • Advanced Design of Experiments in Clinical Investigation. • Clinical Trials • Probability and Stochastic Processes • Computation on Statistics and Optimization. • Advanced Statistical Inference • Multivariate Analysis • Statistical Management Information. • Spatial Epidemiology • Bioinformatics • Survival Analysis
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2014-2015 Master's Degree in Teaching Mathematics (UAB-UB-UPC-UPF-UOC).

2007-2014	Licenciate Degree (5-year plan) in Mathematics in Facultat de Matemàtiques i Estadística (Universitat Politècnica de Catalunya). I did many optional degree credits from the Statistics Degree and the Master's degree of Statistics and Operations research (UB-UPC). • Industrial Statistics • Files and Databases • Generalized Linear Models • Simulation and Queueing Theory
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COMPLEMENTARY COURSES:

- Python for Data Science, AI & Development IBM - Coursera 2024.
<https://www.coursera.org/account/accomplishments/verify/V5Y04W050EPQ>
 - Python project for Data Science
- IBM AI Engineering Specialization - Coursera 2024.
- Introduction to Deep Learning and Neural Networks with Keras - Coursera 2024.
- Applied Machine Learning to Solve Real-World Problems XVII MESIO UPC-UB Summer School 2024.
- Project Management - Barcelona Institute for Global Health (internal course ISGlobal), 2024.
- DataSHIELD Workshop 2019 - Newcastle University.
- Metanalysis with R - MESIO UPC-UB Summer School, 2018.
- Big Data y Estadística - MESIO UPC-UB Summer School, 2015.
- Process Oriented Data Science - MESIO UPC-UB Summer School, 2015.

CORE COMPETENCIES

- **Statistical Engineering:** Advanced Statistical Modelling (GLM/GAM), Multivariate Analysis, and Non-linear regression (Splines/Regularization), Multiple Imputation, Meta Analysis.
- **Advanced Data Science:** End-to-end implementation of Machine Learning pipelines and Deep Learning architectures (CNN, VAE, GNN, PointNet).
- **Generative AI & Computer Vision:** Research and development of Variational Autoencoders (VAE), Latent Space Disentanglement, and Spatial Feature Extraction for high-dimensional data (Medical Imaging).
- **Federated Learning & Privacy:** Expert in conducting federated analysis through DataSHIELD, enabling collaborative research on sensitive datasets without compromising individual privacy or data sovereignty.
- **Analytical Rigor:** Applied Critical Thinking for model validation, hypothesis testing, and rigorous data interpretation in complex environments.

TECHNICAL SKILLS

Programming & Core Data Science

- **Languages:** Python, R (Advanced), C++, SQL (PostgreSQL, Oracle), MATLAB, SAS, STATA.
- **Data Architecture:** MLOps notions (Google Cloud), Federated Analysis (DataSHIELD), Docker.
- **Data Engineering (ETL):** Pandas, NumPy, dplyr/tidyr, MICE (Data Imputation), SQL Developer.

Advanced Modeling & Deep Learning

- **Machine Learning:** Scikit-learn, Caret, glmnet (Regularization), mgcv (GAM/Non-linear).
- **Deep Learning & GenAI:** PyTorch, PyTorch Geometric, TensorFlow, Keras, VAEs (Latent Space).
- **3D & Geometric Learning:** PointNet architectures, Torch Scatter/Cluster, k-NN Graphs.

Visualization & Experiment Tracking

- **Monitoring:** Hyperparameter Tuning, TensorBoard (Real-time), Weights & Biases (notions).
- **Reporting:** Shiny, R Markdown, Jupyter, Plotly, Folium, ggplot2, Matplotlib, Seaborn.

Software Engineering & Workflow

- **DevOps/Collaboration:** Git/GitHub (Version Control), Unit Testing, Markdown, Project Boards.
- **Environments:** Linux, Windows, VS Code, Google Colab, Kaggle API (Large-scale data management).
- **Scientific Publishing:** LaTeX, Microsoft Office Suite (Excel/Access Advanced).

LANGUAGES

Catalan native or bilingual proficiency.

Spanish native or bilingual proficiency.
English: Full professional proficiency (C1 CEFR, CAE).
German: Limited working proficiency.

PUBLICATIONS AND PARTICIPATIONS IN PROJECTS

POSTGRADUATE PROJECT: Flow Matching for Disease Progression in Medical Imaging

Project-AIDL - Final project for the Postgraduate program Artificial Intelligence with Deep Learning 2025–2026, UPC Barcelona

Authors

- Arnau Claramunt
- Sandra Márquez
- Sergi Padrés
- Albert Vidal

Project advisor:

- Òscar Pina

[sanmarquez/Project-AIDL: Final Postgraduate Project: generate-medical-image-disease-progression.](#)

INSTITUTIONAL PROJECTS

Environment and Mother-Child Health/ Chemicals and Water Pollutants - EVOLVE

- **ATHLETE**
Facilitated multi-center environmental epidemiology research by implementing **federated analysis** workflows in **DataSHIELD**, ensuring secure statistical modeling/ Machine Learning analyses across distributed sensitive datasets.
- BiSC
- HELIX
- LIFECYCLE
- ENDOMIX
- INMA
- URBAN KIDS
- ECHOCAT

- EPICEL - Mutua Terrassa (external collaboration).

Climate, Air Pollution, Nature and Urban Health - CANU

- UBDPOLICY
- RESONATE
- iMAP

Collaborations with Ajuntament de Barcelona and Agència de Salut Pública:

- CLIMATE SHELTERS
- ESCOLES VENTILADES
- PROTEGIM LES ESCOLES
- BLOWATER
- ATENCIO
- ECHOCAT: Urban environment and childhood obesity in Catalonia.
- SUPERILLES study in Barcelona.

Past European Projects collaborations or performing analyses for:

- PHENOTYPE
- BlueHealth
- PASTA

Selected recent publications:

Fabbri, L., Andrusaitytė, S., Basagaña, X., Bhopal, S., Bustamante, M., Cheung, R. W., ... & Vrijheid, M. (2025). Prenatal and childhood exposure to mixtures of environmental chemicals and adolescence attentional problems: a triangulation study. *Environment international*, 109927. <https://doi.org/10.1016/j.envint.2023.107856>

Warkentin, S., Fossati, S., Marquez, S., Andersen, A. M. N., Andrusaitytė, S., Avraam, D., ... & Vrijheid, M. (2025). Ambient air pollution and childhood obesity from infancy to late childhood: An individual participant data meta-analysis of 10 European birth cohorts. *Environment international*, 200, 109527. <https://doi.org/10.1016/j.envint.2025.109527>

Warkentin, S., Márquez, S., Vespalcová, H., Knox, B., Gascon, M., Güil-Oumrait, N., ... & Vrijheid, M. (2025). Dietary patterns and exposure to non-persistent endocrine-disrupting chemicals during pregnancy. *Environment International*, 202, 109612. <https://doi.org/10.1016/j.envint.2025.109612>

Sangüesa, J., Márquez, S., Montazeri, P., Fochs, S., Pey, N., Anguita-Ruiz, A., ... & Vrijheid, M. (2025). Role of Maternal Vitamin D3 Levels in Shaping Adolescent Vascular Health: Evidence From a Spanish Population-Based Birth Cohort. *Journal of the American Heart Association*, 14(5), e035273.

<https://doi.org/10.1161/JAHA.124.035273>

López-Toribio, M., Hidalgo, L., Litt, J. S., Daher, C., Nieuwenhuijsen, M., Márquez, S., ... & Ubalde-López, M. (2025). SOOPEN: design and assessment of a tailored systematic observation tool to evaluate outdoor play behavior among schoolchildren groups. *Cities & Health*, 1-12.

<https://doi.org/10.1080/23748834.2024.2439645>

Knox, B., Güil-Oumrait, N., Basagaña, X., Cserbik, D., Dadvand, P., Foraster, M., ... & Vrijheid, M. (2024). Prenatal exposure to per-and polyfluoroalkyl substances, fetoplacental hemodynamics, and fetal growth. *Environment International*, 193, 109090.

<https://doi.org/10.1016/j.envint.2024.109090>

Fernandes, A., Avraam, D., Cadman, T., Dadvand, P., Guxens, M., Binter, A. C., ... & Vrijheid, M. (2024). Green spaces and respiratory, cardiometabolic, and neurodevelopmental outcomes: An individual-participant data meta-analysis of > 35.000 European children. *Environment international*, 190, 108853.

<https://doi.org/10.1016/j.envint.2024.108853>

Domínguez, A., Koch, S., Marquez, S., De Castro, M., Urquiza, J., Evandt, J., ... & Nieuwenhuijsen, M. (2024). Childhood exposure to outdoor air pollution in different microenvironments and cognitive and fine motor function in children from six European cohorts. *Environmental research*, 247, 118174.

<https://doi.org/10.1016/j.envres.2024.118174>